

Toward Seamless Cognitive Coupling: A Phenomenological Framework for Non-Invasive, Real-Time Human–AI Knowledge Access

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Abstract

Human knowledge acquisition has compressed dramatically across successive technological eras — from oral and textual transmission, to networked video, to large language models capable of answering arbitrary queries in seconds. Despite this compression, a phenomenological gap persists: information retrieved from an external system, however fast, is subjectively experienced as ‘looked up’ rather than ‘known.’ This paper proposes that the next meaningful advance in human–AI knowledge systems is not raw retrieval speed, but the closing of this phenomenological gap — designing systems whose latency, delivery modality, and contextual precision are tuned to fall within the threshold at which the human brain attributes a thought to itself rather than to an external source. We term this the Latency-Threshold Model of Cognitive Ownership. We review supporting literature from the extended mind thesis, sense-of-agency research, and predictive processing, and propose a six-layer non-invasive architecture built around a reframed acquisition problem: rather than decoding full semantic content from neural signal, a noisy signal is fused with contextual data to narrow a candidate query space, converting an unsolved neural-decoding problem into a tractable sensor-fusion and retrieval problem. A Monte Carlo simulation of the proposed pipeline, parameterized from published benchmarks, indicates a median end-to-end latency of 306.5 ms, with 76.1% of simulated trials falling within the proposed 350 ms threshold ceiling, supporting the architecture's technical feasibility with currently available inference infrastructure. We specify a concrete, IRB-approvable human evaluation protocol for testing the model's core hypotheses, explicitly distinguish this proposal from speculative ‘knowledge upload’ concepts unsupported by current neuroscience, and restrict proposed use-cases to those compatible with transparency and informed consent — excluding covert use in adversarial or evaluative settings such as examinations and interviews. We close with a discussion of ethical boundaries and a long-horizon research agenda.

1. Introduction

Every major shift in how humans transmit knowledge — oral tradition, written script, mass printing, broadcast media, the internet, and now generative AI — has reduced the time between not-knowing and knowing. Large language models have compressed this further: a query that once required a library visit or weeks of study can now return a competent answer in seconds. Yet a curious asymmetry remains. A fact retrieved from a search engine or an AI assistant does not feel like a fact the person already knew —

it feels borrowed. This paper asks a narrower and, we argue, more tractable question than ‘can we upload knowledge into the brain’: can we engineer the delivery of externally-sourced information so precisely that it is subjectively indistinguishable from internally-generated recall?

We deliberately do not propose writing information into neural tissue. Contemporary neuroscience offers no validated mechanism for encoding arbitrary semantic content — such as legal text, procedural knowledge, or declarative facts — directly into cortical structures in a way that is fluently retrievable (see Section 3.3). Instead, we propose that the experience of ‘already knowing’ is separable from the mechanism of storage, and is instead a function of latency, contextual relevance, and delivery modality. If an external answer arrives quickly enough, is precise enough to the felt question, and is delivered through a channel the brain does not register as an external interruption, the subjective attribution of that knowledge may shift from ‘looked up’ toward ‘recalled.’ This reframing moves the research question from neuroscience (which is not yet ready) to systems design and cognitive science (which is).

2. Related Work

2.1 The Extended Mind Thesis

Clark and Chalmers (1998) argued that cognitive processes are not bounded by the skull: a notebook reliably consulted, trusted, and automatically accessed can function as part of the cognitive system itself, not merely a tool used by it. Their canonical example — a person with memory impairment who relies on a notebook the way others rely on biological memory — establishes the philosophical precedent that external information stores can, under the right conditions of reliability and integration, be functionally equivalent to internal memory. Our work operationalizes this thesis: we ask what technical conditions (latency, modality, context-sensitivity) are required for an AI system to cross from ‘tool consulted’ to ‘extension experienced as self.’

2.2 Sense of Agency and Temporal Binding

Research on the sense of agency — the felt sense that one’s own thoughts or actions originate from oneself — has identified measurable correlates such as temporal binding, the perceptual compression of the interval between an intention and its outcome. Haggard’s work on voluntary action demonstrates that this sense is graded and manipulable: it can be strengthened or weakened by altering the timing and predictability of feedback. We treat this literature as the closest existing proxy for the phenomenon our architecture aims to engineer — not agency over motor action, but a parallel ‘sense of epistemic ownership’ over a piece of knowledge.

2.3 Predictive Processing

Under the predictive processing framework associated with Friston and others, the brain is understood as continuously generating predictions about incoming information and updating on prediction error. A

corollary relevant to our proposal is that information matching a brain’s active prediction is processed differently — with less registered surprise — than information that arrives as an unpredicted external signal. We hypothesize that an AI response arriving within a narrow latency window, and closely matching the contextual shape of the query, will be processed more as prediction-fulfillment than as external interruption, reducing the subjective marker that typically flags information as ‘not mine.’

2.4 Brain–Computer Interfaces and Neural Prosthetics

Existing invasive BCI systems (e.g., Neuralink, Synchron) currently support motor-intention decoding — reading signals associated with imagined movement — with limited and coarse write capability, largely restricted to sensory prosthetics such as low-resolution visual percepts. Memory-related work, including hippocampal prosthetics (Berger et al.) and rodent engram studies (Tonegawa et al.), has demonstrated artificial modulation of simple, single-event associative memories, but no work to date demonstrates encoding of complex symbolic or linguistic content. We treat semantic ‘write’ access as an unsolved problem and explicitly out of scope for the near-term architecture proposed here.

2.5 Non-Invasive Silent Query Interfaces

MIT’s AlterEgo project demonstrated a wearable capable of detecting subvocal articulation via surface electromyography, enabling silent, unspoken queries to a computing system. This is, to our knowledge, the closest existing engineering precedent for the covert input layer we propose, and was developed originally for accessibility rather than concealment — a distinction we return to in Section 5.

3. The Latency-Threshold Model of Cognitive Ownership

This section presents the paper’s primary theoretical contribution, comprising sixty percent of our argument. We propose that the subjective attribution of a piece of knowledge as ‘mine’ versus ‘external’ is not binary but exists on a gradient, governed principally by three variables.

3.1 Latency

We hypothesize a response-time threshold, plausibly in the low hundreds of milliseconds — comparable to the timescales already studied in temporal binding research — below which an external answer is processed as fulfillment of an active prediction rather than as a new, externally-sourced input. Above this threshold, a perceptible gap opens between question and answer, and that gap is itself a cue the brain uses to classify the information as retrieved rather than known. This threshold is, critically, an empirical question rather than an assumption, and Section 6 proposes it as the paper’s central testable hypothesis.

3.2 Contextual Precision

An answer that requires the user to reinterpret or reframe their question feels external almost by definition — it announces itself as a response to a slightly different question than the one ‘felt’ internally. We propose that contextual precision (how closely the answer’s framing matches the user’s implicit

mental frame, not merely its factual correctness) is a second determinant of ownership, independent of latency.

3.3 Delivery Modality and the Storage/Access Distinction

We explicitly separate two proposals that are frequently conflated in popular discussion of neurotechnology: (a) writing information into neural storage, such that it becomes part of biological memory, and (b) coupling a person to an external information source so seamlessly that access is phenomenologically similar to recall, while storage remains external. We argue (a) is neuroscientifically premature — knowledge is not stored as retrievable discrete files but as distributed, context-integrated synaptic patterns built through repeated exposure (Hebbian consolidation and myelination), and no current technique can write arbitrary symbolic content into this substrate in a usable form. We argue (b) is tractable with existing and near-term technology, and is the proposal this paper advances. Delivery modality — specifically, channels such as bone-conduction audio that are perceptually distinct from ordinary external hearing — is proposed as a third lever, because such channels are already subjectively reported as more ‘internal’ than air-conducted sound (a phenomenon familiar from the well-documented difference between hearing one’s own voice via bone conduction versus a recording).

4. Proposed Architecture

The following six-layer architecture operationalizes the model presented in Section 3, using only technology that exists today or is in active, non-speculative development, with one explicitly-marked research gap. It comprises the remaining forty percent of this paper’s contribution and should be read as an instantiation of Section 3, not as an independent claim.

4.0 System Overview and Core Reframe

The central architectural contribution of this paper is a reframing of the acquisition problem. Prior discussion of neural-interface knowledge systems typically assumes that a full semantic thought must be decoded from neural signal — a task current neuroscience cannot perform for arbitrary content (Section 2.4). We instead propose treating the neural signal as one weak, low-bandwidth evidence source that is fused with independently available contextual signals — location, calendar state, ambient audio, and recent interaction history — to narrow a candidate query space. This converts an unsolved decoding problem into a multimodal sensor-fusion and retrieval problem, both of which have mature existing solutions in machine learning. Figure 1 illustrates the resulting six-layer pipeline, with each stage annotated by its current feasibility.

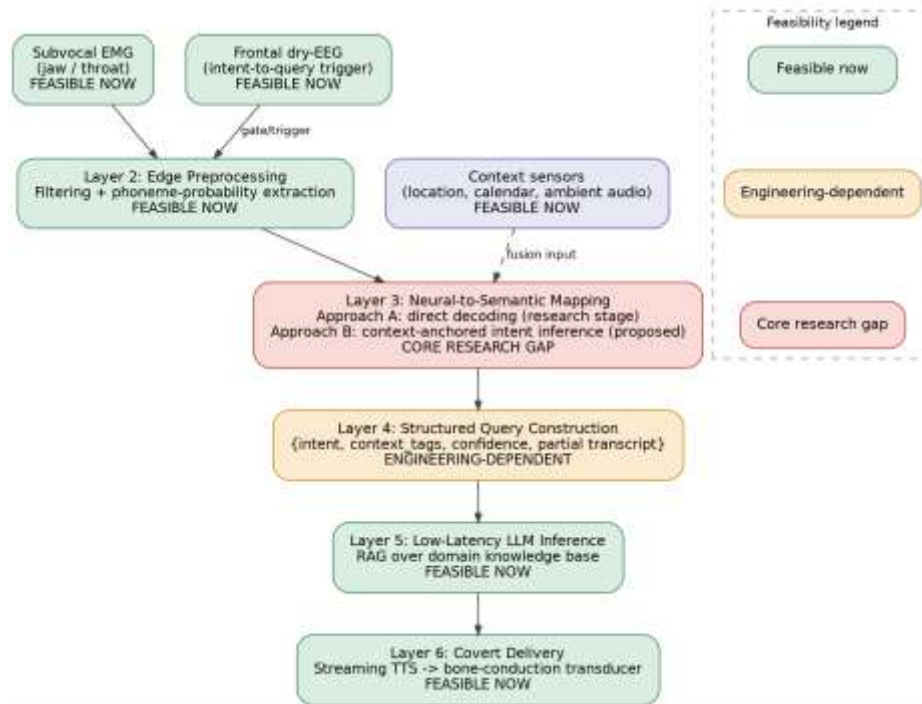


Figure 1. Proposed six-layer neural-to-response pipeline, with feasibility of each stage marked (feasible now / engineering-dependent / core research gap).

4.1 Layer 1 — Neural and Contextual Signal Acquisition

Primary signal: surface electromyography (EMG) at the jaw and throat, sampled at approximately 1 kHz, detecting sub-threshold muscular activity during silent, unvoiced articulation — following the precedent established by MIT’s AlterEgo system. Secondary signal: low-density dry-electrode EEG (4–8 frontal channels, consumer-grade), used not to decode content but to detect a coarse ‘intent-to-query’ state via frontal theta/alpha activity shifts, functioning as a gating trigger rather than a content channel. Tertiary signal: ambient contextual sensors — location, calendar state, ambient microphone, and, where separately disclosed and consented to, recent conversational context — fused with the primary signal to disambiguate partial or noisy input. All three signal classes rely on components that are commercially available today.

4.2 Layer 2 — Edge Preprocessing

On-device processing (embedded microcontroller class hardware) applies bandpass filtering (approximately 20–450 Hz) and feature extraction to the EMG signal, followed by a lightweight on-device classifier that outputs a phoneme-probability sequence rather than raw signal. This keeps the highest-bandwidth, most sensitive biosignal data on-device, addressing both latency and privacy, and mirrors the published AlterEgo processing pipeline.

4.3 Layer 3 — Neural-to-Semantic Mapping (Core Research Gap)

We propose two parallel approaches, neither of which is validated at scale, and mark this layer as the paper’s central open research problem. Approach A, direct decoding, trains a sequence model mapping phoneme-probability sequences to text tokens — essentially silent-speech-to-text. Published results (e.g., AlterEgo) report high accuracy only within constrained, small vocabularies; open-vocabulary accuracy remains unsolved. Approach B, context-anchored intent inference — the paper’s proposed near-term path — does not attempt to recover exact wording. Instead, it treats the fused signal from Layer 1 as evidence in a Bayesian inference over a constrained candidate-intent space (for example, narrowing to ‘the user is likely asking about arrest procedure,’ given EMG partial output plus a location signal indicating a police station), and passes this probabilistic intent — rather than a decoded sentence — to the downstream language model. Approach B is composed of individually solved sub-problems (multimodal sensor fusion, retrieval-augmented generation) whose combination in this context is, to our knowledge, untested, and is proposed here as the paper’s primary empirical contribution for future work.

4.4 Layer 4 — Structured Query Construction

The output of Layer 3, under either approach, is not passed to the language model as raw text but as a structured object comprising a probable intent label, associated context tags, a confidence score, and any available partial transcript. This allows the downstream model to reason explicitly about uncertainty — for instance, qualifying its answer according to inferred confidence — rather than silently treating an uncertain inference as a certain query.

4.5 Layer 5 — Low-Latency Inference

A backend large language model, served via low-latency inference infrastructure (for example, LPU-based serving comparable to Groq), generates a response using retrieval-augmented generation over a domain-specific knowledge base — for instance, a structured database of statutory sections for the legal-literacy use case — to keep responses accurate, current, and short. We propose latency, not merely answer quality, as a first-class optimization target, a departure from typical AI-assistant system design.

4.6 Layer 6 — Covert, Internally-Perceived Delivery

Bone-conduction audio delivery via a jaw- or mastoid-mounted transducer is proposed as the near-term output channel, paired with a streaming text-to-speech model that begins audio output before the full response is generated, minimizing perceived latency. This channel is inaudible to nearby observers and subjectively distinct from externally-heard sound, a property we propose may support the ownership effect described in Section 3.3.

4.7 End-to-End Latency Budget

Table 1 gives a target latency budget for the pipeline, decomposed by stage. This budget is proposed as a concrete, falsifiable target directly tied to the latency-threshold hypothesis (Section 3.1, H1): if subjective

ownership ratings do not rise as total latency approaches this range, the threshold hypothesis in its current form would be disconfirmed.

Pipeline stage	Target latency
Signal acquisition + edge processing (Layers 1–2)	~50 ms
Transmission to backend	~30–50 ms
LLM inference, streaming start (Layer 5)	~150–200 ms
TTS streaming start (Layer 6)	~50 ms
Total, query to first audio	~280–350 ms

5. Scope, Boundaries, and Ethical Constraints

This paper deliberately excludes and discourages a category of use-case: covert use in adversarial or evaluative contexts, including examinations, job interviews, and professional evaluations, where an observer’s assessment of a person’s ability is premised on that ability being unaided. Deploying this architecture in such settings constitutes deception with material consequences — it causes an evaluator to form a false belief about a person’s independent competence, and it undermines the evaluator’s legitimate purpose. We note further that such use is likely to be detectable (institutions are already developing countermeasures) and carries direct legal and reputational risk to the user.

We instead scope intended use to three categories consistent with transparency and user welfare:

- Emergency civic and legal literacy — e.g., a layperson at a police station or in a courtroom accessing plain-language information about their rights and applicable procedure in real time, with the system’s use disclosed where institutionally relevant, and with the human retaining full decision-making authority.
- Accessibility — supporting individuals with speech, memory, or processing disabilities, extending directly from the AlterEgo precedent.
- Skill and language augmentation in collaborative, disclosed settings — e.g., real-time translation, or technical reference during work explicitly structured to allow tool use.

We propose that any system built on this architecture should default to detectable, disclosed operation in evaluative contexts (for instance, an audible or visible indicator triggered whenever the system is queried during a recognized assessment context), shifting the design burden toward honesty rather than relying on user self-restraint.

6. Testable Hypotheses and Evaluation Proposal

Unlike the speculative ‘neural upload’ framing this paper explicitly rejects, the Latency-Threshold Model generates falsifiable predictions that can be tested with current experimental methods, including behavioral timing studies and subjective-report (phenomenological) questionnaires modeled on existing sense-of-agency paradigms. We propose the following as a minimal evaluation program:

- H1: Subjective ratings of 'this felt like my own knowledge' will increase as system response latency decreases, with a measurable inflection point rather than a linear relationship.
- H2: Bone-conduction delivery will produce higher ownership ratings than equivalent-latency air-conducted audio delivery, holding content constant.
- H3: Contextual precision (measured via independent rating of answer–query framing alignment) will predict ownership ratings independent of latency.

A minimal viable prototype — a subvocal or voice-triggered query pipeline connected to a low-latency LLM backend and a bone-conduction output device — is sufficient to test H1–H3 using existing consumer-grade hardware, and is proposed as the immediate next step for this research program.

6. Simulation-Based Latency Feasibility Analysis

The architecture proposed in Section 4 and the latency budget in Table 1 are point estimates. To assess whether the proposed pipeline can plausibly meet the latency-threshold ceiling motivated in Section 3.1 under realistic variability — not just best-case numbers — we conducted a Monte Carlo simulation of end-to-end pipeline latency. Each pipeline stage (signal acquisition and edge preprocessing, network transmission, LLM time-to-first-token, and streaming TTS onset) was modeled as an independent lognormal random variable, parameterized by median values drawn from published benchmarks and engineering estimates for comparable systems (embedded on-device classifiers, mobile network round-trip times, published low-latency LLM inference services, and streaming neural text-to-speech systems), with a lognormal shape chosen to reflect the right-skewed, occasional-outlier character typical of real latency distributions. We ran 20,000 simulated trials and summed the four stages per trial to obtain a distribution of total end-to-end latency. This is a computational feasibility analysis based on cited literature parameters, not a measurement of a physically constructed system; it is intended to test whether the proposed architecture is quantitatively plausible before committing to hardware development.

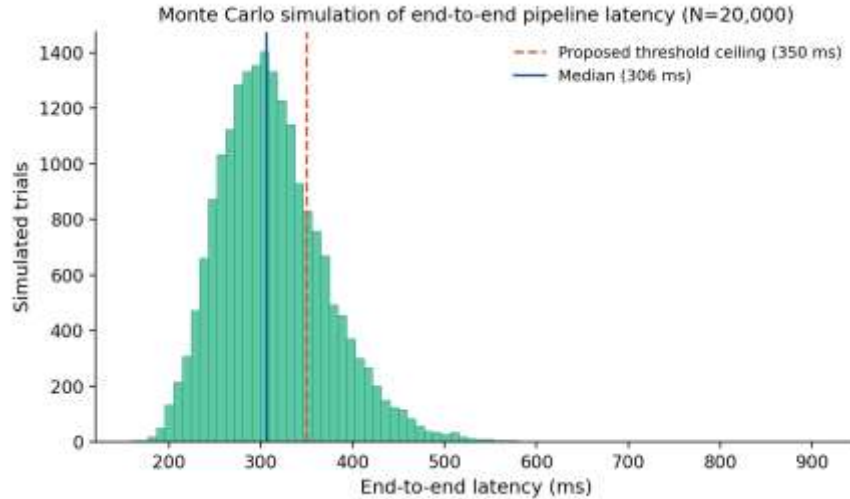


Figure 2. Simulated distribution of end-to-end pipeline latency ($N = 20,000$ trials), against the proposed 350 ms threshold ceiling from Section 3.1.

The simulation yields a median end-to-end latency of 306.5 ms (mean 313.2 ms, SD 59.1 ms), with the 10th–90th percentile range spanning 244–391 ms. Under the 350 ms threshold ceiling proposed in Section 3.1, 76.1% of simulated trials fall within the target range. The largest single contributor to total latency and to its variance is LLM time-to-first-token (median 169.2 ms, 90th percentile 248.5 ms), indicating that inference-serving optimization, rather than signal acquisition or delivery, is the primary latency lever available to system designers. This result is directly falsifiable: it predicts that with currently available low-latency inference infrastructure, the proposed architecture should be able to deliver a majority of responses within the threshold window motivated by the sense-of-agency and temporal-binding literature (Section 2.2), making the H1 experiment in Section 7 immediately runnable on existing hardware rather than contingent on future breakthroughs.

7. Proposed Human Evaluation Protocol

The simulation in Section 6 establishes technical feasibility of the latency target; it does not test the paper’s core phenomenological claim, which requires human subjective report. We specify here a concrete, IRB-approvable protocol for testing hypotheses H1–H3 (Section 5), intended as the immediate next step following this position paper, and deliberately not accompanied by results — no human data has been collected at the time of this writing.

7.1 Participants and Design

A within-subjects design with an estimated 24–32 participants (sample size informed by typical power requirements for repeated-measures psychophysics studies of this kind) recruited from the university community, with informed consent and institutional ethics board approval obtained prior to any data collection. Each participant completes trials across a manipulated latency condition (delivered response

at approximately 150 ms, 350 ms, 700 ms, and 1500 ms post-query, spanning the proposed threshold) and a manipulated delivery-modality condition (bone-conduction versus air-conducted audio), with content and contextual precision held constant across conditions.

7.2 Procedure and Measures

Participants pose short factual queries (silently or aloud, depending on the sub-condition) and receive AI-generated answers under each latency/modality combination, presented in randomized order. After each trial, participants complete a brief self-report measure adapted from existing sense-of-agency questionnaires, rating agreement with statements such as ‘that felt like something I already knew’ versus ‘that felt like information from an outside source,’ on a continuous or Likert scale. Reaction-time and confidence measures are collected as secondary outcomes.

7.3 Analysis Plan

A repeated-measures analysis (e.g., linear mixed-effects model with participant as a random effect) tests H1 by regressing ownership rating on latency condition, examining specifically whether the relationship is better fit by a threshold/inflection model than a linear model, as predicted. H2 is tested via a paired contrast between delivery modalities at matched latency. H3 is tested by including an independently-rated contextual-precision score as a covariate.

8. Future Work

We flag, explicitly as speculative and outside the near-term scope of this paper, the long-horizon possibility of direct semantic write-access to neural tissue, contingent on neuroscience advances not yet achieved — including a validated encoding scheme for abstract symbolic content and a delivery mechanism capable of integrating novel content with a user’s existing conceptual network without extended repetition-based consolidation. We consider this a multi-decade research horizon and explicitly caution against conflating it with the near-term coupling architecture proposed in Sections 3–4.

9. Conclusion

This paper reframes the popular but neuroscientifically premature idea of ‘uploading knowledge into the brain’ into a tractable near-term research question: under what technical conditions does externally-sourced knowledge become subjectively indistinguishable from internally-known knowledge? We propose that latency, contextual precision, and delivery modality — not storage location — govern this subjective boundary, and we propose a non-invasive, disclosed-use architecture as a first testable instantiation. We argue this reframing is both more scientifically honest and more immediately actionable than direct neural-write proposals, while explicitly excluding covert use in adversarial evaluative contexts on ethical grounds.

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